

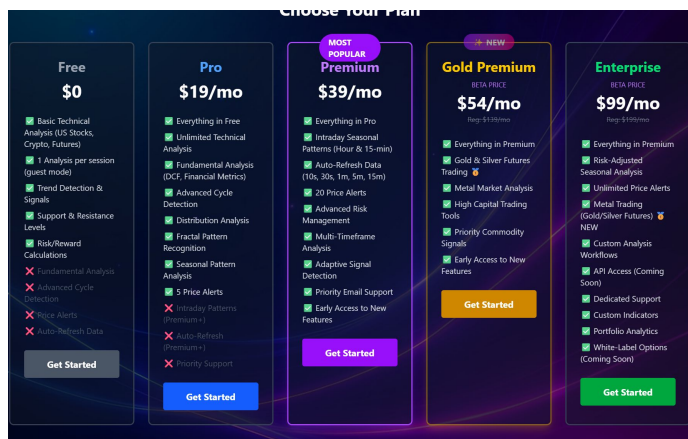
Stock-AI Demo #1

Team UG_HF_03

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Problem

- Stock trading is complicated
- Learning the market is difficult
- Platforms with learning focus exist, but are inaccessible
 - Use proprietary LLMs
 - No free service, often very expensive



Solution

- Use AI to provide intuitive & interactive help with trading
- Make our AI-integrated platform free & accessible

Two-pronged development

- Need to build AI Agent capable of helping users
- Need stock trading platform nuanced enough to integrate & evaluate AI

Team task breakdown

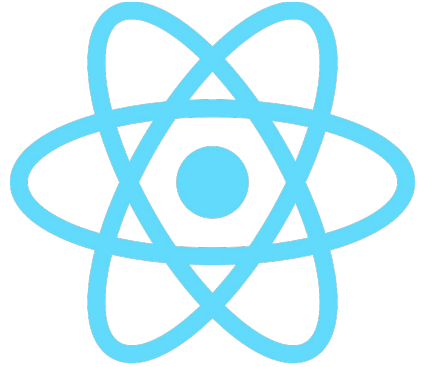
- Graduate Team
 - Building & refining open-source model for stock trading
 - Produce thoughts/recommendations on a given trade
 - Chat with model
- Undergraduate Team (us)
 - Create web platform for mock stock trading
 - User accounts with portfolios, ability to inspect market & make mock trades
 - Integrate features of LLM into user experience

Development problems

- Real-time data access
 - Solution: yfinance websocket, free & unlimited use
- Secure user accounts + user actions
 - Hashed password storage + JSON Web Tokens (JWTs)
- Backend downtime & Deployment issues
 - Prevented frontend-backend connection
 - Containerization of backend to streamline deployment

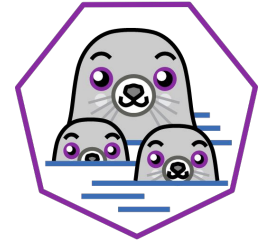
Technologies in use - Frontend

- **React.js**
 - Web framework
 - Templates & automatic HTML generation
- **Lightweight Charts**
 - Visualizing stock data
- **Material UI**
 - Component Library



Technologies in use - Backend

- FastAPI Python framework
 - Easy creation of endpoints
 - Automatic Jagger documentation
- MariaDB
 - MySQL database
- Docker / Podman
 - Local & server containerization, respectively
- Insomnia / Postman
 - Testing endpoints



Project Status

- User Authentication
- Protected Routes

STOCK-AI API openapi.com GET OAuth 2.0

This is the API for Stock-AI, an AI-integrated stock trading platform. Currently a work in progress, endpoints & functionality are subject to change.

[Authorize](#)

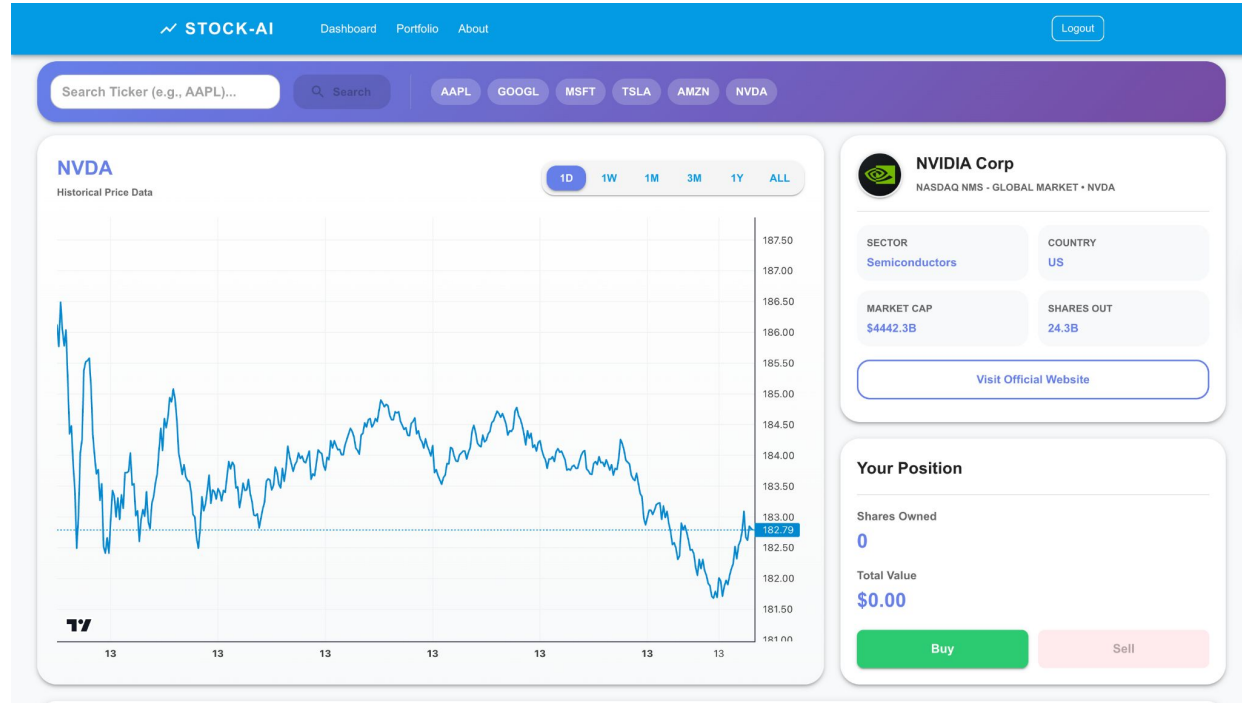
users		^
POST	<code>/users/create</code>	User account creation / signup
POST	<code>/users/login</code>	User account login
GET	<code>/users/protected</code>	Test JWT validity
portfolios		^
POST	<code>/portfolios/create</code>	Create a new portfolio for user
GET	<code>/portfolios/listall</code>	Grab all portfolios for user
data		^
GET	<code>/data/{ticker}/profile</code>	Read Stock Profile
GET	<code>/data/{ticker}/history</code>	Get Stock History
GET	<code>/data/{ticker}/news</code>	Get Stock News Endpoint
default		^
GET	<code>/</code>	root request
GET	<code>/num_users</code>	Grabs number of users listed in database

Login Page

[Login](#)[Register](#)

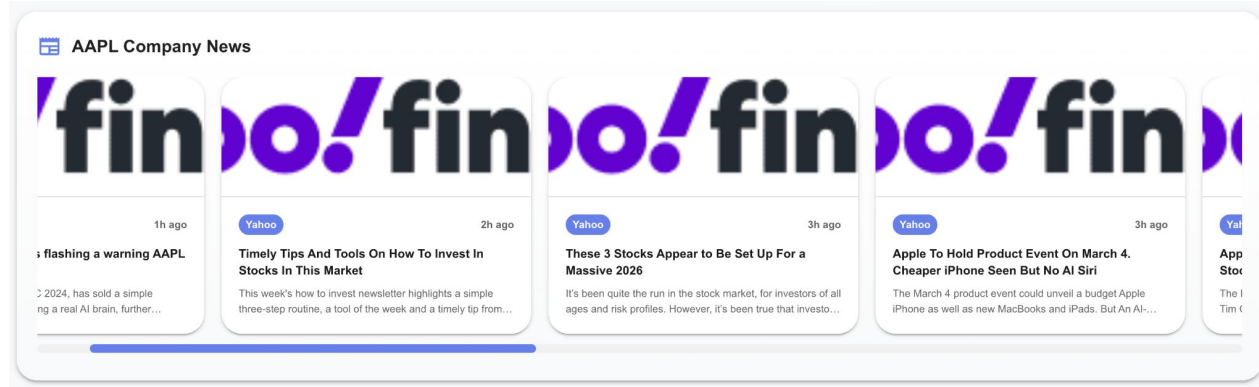
Project Status

- Stock dashboard
 - Ticker lookup
 - Historical prices
 - Company profile
 - Buy/Sell positions



Project Status

- Stock dashboard
 - Company News



Project Status

- Stock Portfolio

My Portfolios

[Create](#)

Test (Created: 2/16/2026)

Project Status

- About Page

A Tool to Help Anyone Practice Investing Stocks

Requested by Dr Hung D Phan, this website is a senior design project developed to help lorem ipsum stuff and stuff.



All
Stocks Supported

0
AI Features

Quick Demo

What's next?

- Expanding on portfolio functionality
 - Allow users to buy/sell
- AI Integrations
 - Give users recommendations

Challenges

- AI Integration Ambiguity
- Error handling
- Communication

Timeline for remaining features

- Demo 2
 - Features
 - Watchlist, Cash Balance, Transaction History, More Detailed Portfolio
 - This Week
 - Plan tickets and responsibilities
- Demo 3
 - Features
 - AI Chat features and suggestions
 - Responsive UI that deals with AI Suggestions